Daehee Park

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RESEARCH INTERESTS

Deep learning
- Multi-modal deep learning and graph neural networks

Trajectory prediction
- Predicting future motion of various agents under interacting system and its applications

Transfer learning
- Improving adaptability of deep learning model on unseen data during training/test time

Computer vision
- Perception methods including detection, tracking and segmentation

EDUCATION

KAIST, Daejeon, Korea 2020.03 - 2025.02(expected)
Ph.D. in Mechanical Engineering
Advisor: Kuk-Jin Yoon

KAIST, Daejeon, Korea 2018.03 – 2020.02
M.S. in Mechanical Engineering
Thesis: Removal of reflected virtual images in visual recognition utilizing 3D depth information

KAIST, Daejeon, Korea 2013.03 – 2018.02
B.S. in Mechanical Engineering (Double major in Business and Technology Management)

Gwangju Science High School for the Gifted, Gwangju, Korea 2011.03 – 2013.02

PUBLICATIONS

(*: equal contribution)

[c8] T4P: Test-Time Training of Trajectory Prediction via Masked Autoencoder and Actor-specific Token Memory
Daehee Park, Jae-Seok Jeong, Sung-Hoon Yoon, Jaewoo Jeong, Kuk-Jin Yoon
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024

[c7] Multi-agent Long-term 3D Human Pose Forecasting via Interaction-aware Trajectory Conditioning
Jaewoo Jeong*, **Daehee Park***, Kuk-Jin Yoon
Conference on Computer Vision and Pattern Recognition (**CVPR**), **Highlight**, 2024
Top 2.81% of all papers

[c6] Diffusion-Guided Weakly Supervised Semantic Segmentation
Sung-Hoon Yoon, Hoyong Kwon*, Jaeseok Jeong*, **Daehee Park**, and Kuk-Jin Yoon
European Conference on Computer Vision (**ECCV**), 2024

[c5] Improving Transferability for Cross-domain Trajectory Prediction via Neural Stochastic Differential Equation
Daehee Park, Jaewoo Jeong, Kuk-Jin Yoon
AAAI Conference on Artificial Intelligence (**AAAI**), 2024

[c4] Leveraging Future Relationship Reasoning for Vehicle Trajectory Prediction
Daehee Park, Hobin Ryu, Yunseo Yang, Jegyeong Cho, Jiwon Kim, Kuk-Jin Yoon
International Conference on Learning Representations (**ICLR**), 2023
1st rank in nuScenes prediction challenge

[c3] BIPS: Bi-modal Indoor Panorama Synthesis via Residual Depth-Aided Adversarial Learning
Changgyoon Oh*, Wonjune Cho*, Yujeong Chae*, **Daehee Park***, Lin Wang, Kuk-Jin Yoon
European Conference on Computer Vision (**ECCV**), 2022

[c2] Unlocking the potential of ordinary classifier: Class-specific adversarial erasing framework for weakly supervised semantic segmentation
Hyeokjun Kweon*, Sung-Hoon Yoon*, Hyeonseong Kim, **Daehee Park**, Kuk-Jin Yoon
Proceedings of the IEEE/CVF international conference on computer vision (**ICCV**), 2021

[j1,c1] Identifying Reflected Images from Object Detector in Indoor Environment Utilizing Depth Information
Daehee Park, Yong-Hwa Park
IEEE Robotics and Automation Letters (**RA-L**) & IEEE International Conference on Robotics and Automation (**ICRA**), 2020

WORK EXPERIENCE

Qualcomm (Deep Learning R&D Intern) Remote, 2024.04 – 2024.09

- Work with the US team in remote
- Development of deep learning models to predict trajectories and/or intentions for road users

Naver Labs, (Research Intern) Seongnam, Korea, 2021.06 – 2021.08

- Developing deep learning model that predict future trajectory of road agents for autonomous driving
- Development of realistic driving simulator using deep trajectory prediction network

PATENT

Electronic device and method of identifying false image of object attributable to reflection in indoor environment thereof
Yong-hwa Park, **Daehee Park**, Simeneh Semegne Gulelat
US-Registration No. US11282178B2 / EP-Registration No. EP3772701 / KR-Registration No. 10-2287478-0000

PROJECTS

Development of identifying abnormal vehicles under complex road scenario, 2021.03 – 2024.03
Funded by Electronics and Telecommunications Research Institute (ETRI)

- Designing, building and implementing deep neural network for abnormal vehicle detection in video
- Developing robust prediction system against perception error in video using Neural SDE
- Publishing a paper to top-level machine learning conference (AAAI)

Development of trajectory prediction method of road agents for autonomous driving, 2020.09 – 2022.12
Funded by Naver Labs

- Designing and building state-of-the-art trajectory prediction network for autonomous driving.
- Improving long-term prediction under multi-agent interacting system
- Publishing a paper to top-level machine-learning conference (ICLR)

Development of rescue boat technology for coast guard, 2019.09 – 2020.08
Funded by Korea Institute of Marine Science and Technology (KIMST)

- Designing, development of vision system and specifying sensor
- Developing small target detection method using thermal (IR) imaging

Development of human-care robot technology for aging society, 2018.03 – 2020.03
Funded by Electronics and Telecommunications Research Institute (ETRI)

- Designing, building and implementing human detection and tracking algorithm
- Developing personalized human tracker for service robots via online learning
- Publishing a paper to top-level robotics conference and journal (ICRA & RA-L)

AWARD AND HONORS

- Research Scholarship, Funded by Samsung Advanced Institute of Technology (2022.09~)
- 1st Place in Best Paper Awards during IPIU 2022 (2022.02)
- Outstanding Achievement Award for Academic Excellence, KAIST ME (2016)

ACADEMIC SERVICE

- **Reviewer of robotics conferences & journal**
  RA-L, ICRA, IROS, T-IV
- **Reviewer of ML&CV conferences**
  NeurIPS, ICLR, CVPR, ECCV, CVPR

COMPUTER SKILLS

- **Programming Language**
  Strong: Python, MATLAB
  Knowledgeable: C, C++
- **Open Source**
  Pytorch, Ray, Git, PyG, Carla Simulator

LANGUAGES

Korean, English